

Scalability & Future Plan

Date	25 September 2026
Team ID	6ab22227a614981d6d2c6451
Project Name	Import Data using Transform Maps
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Excel file size limits	Very large datasets (>100k rows) might timeout	Medium
2	Manual upload dependency	Requires admin to manually click Load Data	High

Scalability Plan:

S.No	S.No Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	5-10 Admins	Scale out API nodes if concurrent uploads increase
2	Data Storage	Relational DB	Archive old import sets regularly
3	Performance	Batch processing	Increase batch size limit in Transform Engine
4	Security	Role-based access	Integrate with corporate SSO

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Automated SFTP data ingestion	Q1 2027	Eliminates manual Excel upload
Phase 3	Machine Learning anomaly detection	Q3 2027	Flags corrupted data before insert
Phase 4	Real-time API integration with HRIS	Q1 2028	True real-time sync